[WIP] DE Schema Structure for Supply Planning Web API

Overview:

This document defines the initial database schema design to support the Supply Planning Web API.

[Lucid link for design schema]

Tables

1. `supply_planning.vanning_center`

This table provides information about the available vanning centers.

MVP - User does not get an ability to change/add vanning centers, this needs to be done through backend.

Column name	Data Type	Description	Constraints
vc_id	UUID	Unique identifier for each plant line	Primary Key
vanning_center	TEXT	Vanning center mapped to specific NAMC line.	Not Null
is_active	BOOL	Indicates whether the vanning center is active.	Not Null
effective_in	DATE	Date when the vanning center becomes effective.	Not Null
effective_out	DATE	Date when the vanning center	Nullable

		becomes ineffectiveve.	
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp without time zone
last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE SCHEMA IF NOT EXISTS supply_planning;

CREATE TABLE IF NOT EXISTS supply_planning.vanning_center (
    vc_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    vanning_center TEXT NOT NULL,
    is_active BOOLEAN NOT NULL,
    effective_in DATE NOT NULL,
    effective_out DATE,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT
);
```

2. `supply_planning.namc_line_subseries`

This table consolidates production site information and the associated vehicle sub-series managed by each NAMC line.

Column name	Data Type	Description	Constraints

namc_line_series_id	UUID	Unique identifier for each plant line	Primary Key
namc	TEXT	Name of the namc	Not Null
line	TEXT	Name of the namc line	Not Null
idline	INTEGER	Line id	Not Null
sub_series_description	TEXT	Sub-series name	Not Null
sub_series	VARCHAR(2)	sub-series code	Not Null
ssn	VARCHAR(2)	Series	Not Null
ssn_description	TEXT	Series name	Not Null
is_active	BOOL	Indicates whether the mapping is active.	Not Null, Default True
global_rundown_key	TEXT	Unique code for each Name	Nullable
effective_in	DATE	Date when the NAMC line and subseries mapping becomes effective.	Not Null
effective_out	DATE	Date when the NAMC line and subseries mapping becomes ineffective.	Nullable

created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp without time zone
last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.namc_line_subseries (
    namc_line_series_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    namc TEXT NOT NULL ,
    line TEXT NOT NULL ,
    idline INTEGER NOT NULL,
    sub_series_description TEXT NOT NULL,
    sub_series VARCHAR(2) NOT NULL,
    ssn VARCHAR(2) NOT NULL,
    ssn_description TEXT NOT NULL,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    global_rundown_key TEXT,
    effective_in DATE NOT NULL,
    effective_out DATE,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT
);
```

3. `supply_planning.user_details`

This table stores details of all users who will interact with the supply planning application. It captures essential user information and their associated default plants.

Column name	Data Type	Description	Constraints

user_id	UUID	Unique identifier for each user	Primary Key
user_email	TEXT	User email	Not Null, Unique
user_name	TEXT	Full name of the user	Not Null
default_config	TEXT	List of default plants associated with the user	Nullable
created_date_timestamp	TIMESTAMP	Timestamp when the user was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the user was last updated	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.user_details (
    user_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_email TEXT UNIQUE,
    user_name TEXT NOT NULL,
    default_config TEXT,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE
);
```

4. `supply_planning.scenarios`

This table stores all planning scenarios created by planners. It captures scenario metadata and status, which supports the landing page and scenario management in the web UI.

Column name	Data Type	Description	Constraints
scenario_id	UUID	Unique identifier for each scenario	Primary Key
user_name	TEXT	Reference to the user who created	Not Null

		the scenario	
user_email	TEXT	Reference to the user email who created the scenario	Foreign Key (`user_details.user_email`), Not Null
scenario_name	TEXT	Name of the scenario	Not Null, Unique
namc	TEXT	Plant associated with the scenario	Not Null
line	TEXT	Production line associated with the scenario	Not Null
plan_type	TEXT	Type of plan (e.g., ap, getsudo)	Not Null, check plan_type in ('AP', 'Getsudo')
scenario_cycle	TEXT	scenario cycle start date(Jan26)	Not Null
start_month_year	TEXT	Start date of the planning period(YYYYMM)	Not Null
end_month_year	TEXT	end date of the planning period(YYYYMM)	Not Null
getsudo_month	TEXT	Start month for **getsudo** cycle, can be set to null in case of **ap** (YYYYMM)	Nullable
ap_month	TEXT	Start month for AP (YYYYMM)	Nullable

scenario_status	TEXT	Current status of the scenario (Not Started, In Progress, Completed)	Not Null
is_active	BOOLEAN	Shows scenario is active or not	Not Null, Default True
created_date_timestamp	TIMESTAMP	Timestamp when the scenario was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the scenario was last updated	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.scenarios (
    scenario_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_email TEXT NOT NULL,
    user_name TEXT NOT NULL,
    scenario_name TEXT NOT NULL,
    namc TEXT NOT NULL,
    line TEXT NOT NULL,
    plan_type TEXT NOT NULL CHECK (plan_type IN ('Getsudo', 'AP'),
    scenario_cycle TEXT NOT NULL,
    start_month_year TEXT NOT NULL,
    end_month_year TEXT NOT NULL,
    getsudo_month TEXT,
    ap_month TEXT,
    scenario_status TEXT NOT NULL DEFAULT 'Not Started',
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    CONSTRAINT fk_user FOREIGN KEY (user_email) REFERENCES supply_planning.user_details(user_email),
    CONSTRAINT chk_scenario_status CHECK (scenario_status IN ('Not Started', 'In Progress', 'Completed'))
);
```

5. `supply_planning.namc_production_calendar`

The `namc_production_calendar` table records the production details for NAMC sites and lines. It tracks daily production status, including working days, holidays, and workday percentages, supporting production planning and scheduling.

Column name	Data Type	Description	Constraints

namc_prd_cal_id	UUID	Unique identifier for each record	Primary key
scenario_id	UUID	Reference to the associated planning scenario	Foreign Key (`scenarios.scenario_id`), Not Null
namc	TEXT	Identifier for the NAMC site	Not Null
line	TEXT	Production line identifier	Not Null
prod_date	DATE	Calendar date (format: YYYY-MM-DD)	Not Null
week_number	TEXT	Define the week to date mapping. WK202601	Not Null
month_number	INTEGER	Define the month to date mapping. (202601)	Not Null
day_of_week	TEXT	Name of the day of the week (e.g., Monday)	Not Null
is_working_day	BOOLEAN	Indicates if the date is a working day	Not Null
work_day_percentage	INTEGER	Percentage of the day considered as working time	Not Null, Check (0 <= value <= 200)
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp

last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.namc_production_calendar (
    namc_prd_cal_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    namc TEXT NOT NULL,
    line TEXT NOT NULL,
    prod_date DATE NOT NULL,
    week_number TEXT NOT NULL,
    month_number INTEGET NOT NULL,
    day_of_week TEXT NOT NULL,
    is_working_day BOOLEAN NOT NULL,
    work_day_percentage INTEGER NOT NULL CHECK (work_day_percentage >= 0 AND work_day_percentage <= 200),
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id)
);
```

6. `supply_planning.model_change_date` **(Manual input)**

This table stores information about the model year lifecycle for each sub-series within a given planning scenario. It tracks the start and end dates of each model year.

Column name	Data Type	Description	Constraints
mcd_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated	Foreign Key (`scenarios.sc`

		planning scenario	`enario_id`), Not Null
sub_series_description	TEXT	Vehicle sub-series classification	Not Null
model_year	TEXT	Model year (e.g., - MY 26)	Not Null
start_prod_date	DATE	Start date of the model year (format: YYYY-MM-DD)	Not Null
end_prod_date	DATE	End date of the model year (format: YYYY-MM-DD)	Not Null
created_date_timestamp	TIMESTAMP	When the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	When the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.model_change_date (
    mcd_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    sub_series_description TEXT NOT NULL,
    model_year TEXT NOT NULL,
    start_prod_date DATE NOT NULL,
    end_prod_date DATE NOT NULL,
```

```sql
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id)
);
```

7. `supply_planning.namc_allocation_plan`

This table stores the volume of vehicles planned for production on each **working day**.

Regardless of whether the input originates from **Getsudo (daily-level input)** or **Annual Plan (monthly-level input)**, the final data is persisted at a **daily granularity**. Monthly inputs are first distributed across applicable working days (based on the NAMC calendar) and then stored in the database as daily records.

Column name	Data Type	Description	Constraints
namc_allocation_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated planning scenario	Foreign Key (`scenario.scenario_id`), Not Null
getsudo_month	VARCHAR(7)	(YYYY-MM)	Not Null
namc	VARCHAR(5)	Name of the namc line	Not Null
idline	VARCHAR(2)	Line ID	Not NUll
sub_series_description	TEXT	Vehicle sub-series classification	Not Null
production_date	DATE	Allocation date	Not Null
production_month	VARCHAR(7)	Production month (YYYY-MM)	Not Null

daily_volume	INTEGER	Number of vehicles to be produced on daily level.	Not Null, Check (volume >= 0)
monthly_volume	INTEGER	Number of vehicles to be produced on monthly level.	Not Null, Check (volume >= 0)
ssn	VARCHAR(2)	Series	Not Null
editable	BOOLEAN	Set to **FALSE** when the data is loaded via integration or file upload. Set to **TRUE** when the values are manually edited by the user.	Not Null
created_date_timestamp	TIMESTAMP	When the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	When the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```
CREATE TABLE IF NOT EXISTS supply_planning.namc_allocation_plan (
```

```sql
    namc_allocation_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    getsudo_month VARCHAR(7) NOT NULL,
    namc VARCHAR(5) NOT NULL,
    idline VARCHAR(2) NOT NULL,
    sub_series_description TEXT NOT NULL,
    production_date DATE NOT NULL,
    production_month VARCHAR(7) NOT NULL,
    daily_volume INTEGER NOT NULL CHECK (daily_volume >= 0),
    monthly_volume INTEGER NOT NULL CHECK (monthly_volume >= 0),
    ssn VARCHAR(2) NOT NULL,
    editable BOOLEAN NOT NULL,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id)
);
```

8. `supply_planning.tmc_working_day_calendar`

Column name	Data Type	Description	Constraints
tmc_working_cal_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated planning scenario	Foreign Key (`scenarios.scenario_id`), Not Null
vanning_center	TEXT	Vanning center name	Not Null
vc_date	DATE	Calendar date (format: YYYY-MM-DD)	Not Null
week_number	VARCHAR(8)	WK202601	Not Null
month_number	INTEGER	(YYYY-MM)	Not Null
day_of_week	TEXT	Name of the day of the week (e.g., Monday)	Not Null

is_working_day	BOOLEAN	Indicates if the date is a working day	Not Null
work_day_percentage	INTEGER	Percentage of the day considered as working time	Not Null, Check (0 <= value <= 200)
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.tmc_working_day_cal (
    tmc_working_cal_id PRIMARY KEY DEFAULT gen_random_uuid(),
    vanning_center TEXT NOT NULL,
    scenario_id UUID NOT NULL,
    date DATE NOT NULL,
    week_number VARCHAR(8) NOT NULL,
    month_number INTEGER NOT NULL,
    day_of_week TEXT NOT NULL,
    is_working_day BOOLEAN NOT NULL,
    work_day_percentage INTEGER NOT NULL CHECK (work_day_percentage >= 0 AND work_day_percentage <= 200),
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id) ON DELETE CASCADE
);
```

9. `supply_planning.shipping_pattern`

This table captures the shipping process timeline for parts in the NAMC supply chain. It tracks key dates including:

- **Vanning Date:** When parts are loaded into containers at the vanning center.
- **Vessel Load Date:** When containers are loaded onto the ship.
- **Plant Arrival Date:** When parts arrive at the NAMC plant.

Column name	Data Type	Description	Constraints
sp_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated planning scenario	Foreign Key (`scenarios.scenario_id`), Not Null
namc	TEXT	Plant associated with the shipment	Not Null
vanning_center	TEXT	Location where parts are loaded into containers	Not Null
vanning_date	DATE	Date parts were loaded into containers	Not Null
vanning_day	TEXT	Day of a week	Not Null
tmc_working	BOOLEAN	Indicates if TMC was working on the vanning date	Not Null
carrier	TEXT	Shipping carrier name	Nullable
vessel_load_date	DATE	Date containers were loaded onto the vessel	Not Null

namc_arrival_date	DATE	Date parts arrived at the NAMC	Not Null
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.shipping_pattern (
    sp_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    namc TEXT NOT NULL,
    vanning_center TEXT NOT NULL,
    vanning_date DATE NOT NULL,
    vanning_day TEXT,
    tmc_working BOOLEAN,
    carrier TEXT,
    vessel_load_date DATE,
    namc_arrival_date DATE,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_sp_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id)
);
```

10. `supply_planning.generic_pattern`

This table stores generic shipping patterns. It can be used to generate missing or complete time frames based on the stored patterns.

Column name	Data Type	Description	Constraints
gsp_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated planning scenario	Foreign Key (`scenarios.scenario_id`), Not Null
vanning_day	TEXT	Day on which vanning will happen.	Not Null
vessel_to_namc_days	TEXT	Day on which vessel is loaded to ship	Not Null
namc_arrival_day	INTEGER	No of days between vessel load day to namc arrival days	Not Null
apply_to_all	BOOLEAN	**True**: Entire plan duration **False**: All months outside uploaded coverage	Not Null
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp

last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.generic_pattern (
    gsp_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    vanning_day TEXT NOT NULL,
    vessel_load_day TEXT NOT NULL,
    vessel_to_namc_days INTEGER NOT NULL,
    apply_to_all BOOLEAN NOT NULL,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_gp_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id)
);
```

11. `supply_planning.vanning_lead_time`

This table tracks the lead time between the order date and the vanning date for parts at a specific vanning center.

Column name	Data Type	Description	Constraints
vlt_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated planning scenario	Foreign Key (`scenarios.scenario_id`), Not Null

vanning_center	TEXT	Location where parts are loaded into containers	Not Null
order_date	DATE	Date when the order was placed. (YYYY-MM-DD)	Not Null
order_day	TEXT	Day of a week	Not Null
vanning_date	DATE	Date when parts were loaded into containers(YYYY-MM-DD)	Not Null
vanning_day	TEXT	Day of a week	Not Null
lead_time	INTEGER	No of days between vanning date and order date	Not Null
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE supply_planning.vanning_lead_time (
    vlt_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),,
    scenario_id INT NOT NULL,
    vanning_center TEXT NOT NULL,
    order_date DATE NOT NULL,
    order_day TEXT NOT NULL,
    vanning_date DATE NOT NULL,
    vanning_day TEXT NOT NULL,
    lead_time INTEGER NOT NULL,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_vlt_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id)
);
```

12. `supply_planning.standard_lead_time`

This table stores standard lead time. It can be used to generate missing or complete time frames based on the stored patterns.

Column name	Data Type	Description	Constraints
slt_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated planning scenario	Foreign Key (`scenarios.scenario_id`), Not Null
lead_time	INTEGER	No of days between order date and vanning date.	Not Null
apply_to_all	BOOLEAN	**True**: Entire plan duration **False**: All months outside	Not Null

		uploaded coverage	
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.standard_lead_time (
    slt_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    lead_time INTEGER NOT NULL,
    apply_to_all BOOLEAN NOT NULL,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_slt_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id)
);
```

13. `supply_planning.grouping`

The `grouping` organizes vehicle sub-series into groups based on their associated vanning center within a specific scenario.

Column name	Data Type	Description	Constraints
group_id	UUID	Unique identifier for each record	Primary Key

group_name	TEXT	Name of the group.	Not Null
vanning_center	TEXT	Vanning center associated with the group	Not Null
is_active	BOOLEAN	shows group active status	Default True
effective_in	DATE	Date when group was created.	Not Null
effective_out	DATE		Nullable
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.grouping (
    group_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    group_name TEXT NOT NULL,
    vanning_center TEXT NOT NULL,
    effective_in DATE NOT NULL,
    effective_out DATE,
    is_active BOOLEAN DEFAULT TRUE,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_group_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id) ON DELETE CASCADE
```

```
);
```

14. `supply_planning.group_namc_line_series_mapper`

This table provides details about the namc, line and series that belong to a particular group.

Column name	Data Type	Description	Constraints
group_namc_line_series_id	UUID	Unique identifier for each record	Primary Key
namc_line_series_id	UUID	Reference to the associated namc, line and series	Foreign Key (`namc_line_subseries.namc_line_series_id`), Not Null
group_id	UUID	Reference to the group	Foreign Key (`grouping.group_id`), Not Null
is_active	BOOLEAN	shows mapping is still valid or not	Not null, Default True
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who	Nullable

		updated the record	

```sql
CREATE TABLE IF NOT EXISTS
supply_planning.group_namc_line_series_mapper (
    group_namc_line_series_id UUID PRIMARY KEY DEFAULT
gen_random_uuid(),
    namc_line_series_id UUID NOT NULL,
    group_id UUID NOT NULL,
    is_active BOOLEAN DEFAULT TRUE,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL
DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_mapper_group FOREIGN KEY (group_id) REFERENCES
supply_planning.grouping(group_id),
    CONSTRAINT fk_mapper_namc_line_subseries FOREIGN KEY
(namc_line_series_id) REFERENCES
supply_planning.namc_line_subseries(namc_line_series_id)
)
```

15. `supply_planning.group_scenario_mapper`

This table maps groups to their corresponding scenarios.

Column name	Data Type	Description	Constraints
grp_scenario_mp_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated scenario_id	Foreign Key (`scenarios.scenario_id`), Not Null
group_id	UUID	Reference to the group	Foreign Key (`grouping.group_id`), Not Null
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was	Nullable

		last updated	
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.group_scenario_mapper (
    grp_scenario_mp_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    group_id UUID NOT NULL,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_group_scenario_mapper_group FOREIGN KEY (group_id) REFERENCES supply_planning.grouping(group_id),
    CONSTRAINT fk_group_scenario_mapper_scenarios FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id)
)
```

16. `supply_planning.min_max_doh`

This table stores inventory control information specifying the minimum and maximum Days on Hand (DOH) that must be maintained for each sub series within a vanning center. This helps ensure optimal inventory levels are kept meeting demand while avoiding overstocking.

Column name	Data Type	Description	Constraints
min_max_doh_id	UUID	Unique identifier for each record	Primary Key
group_id	UUID	Reference to the associated group	Foreign Key (`grouping.group_id`), Not Null
scenario_id	UUID	Reference to the associated	Foreign Key (`scenarios.sc`

		planning scenario	`enario_id`), Not Null
month	INTEGER	month for which the DOH limits apply	Not Null
year	INTEGER	Year for which the particular month DOH limits apply	Not Null
min_doh	INTEGER	Minimum Days On Hand inventory level to maintain	Not Null
max_doh	INTEGER	Maximum Days On Hand inventory level to maintain	Not Null
apply_to_all_months	BOOLEAN	Shows status of toggle button on UI	Not Null, Default TRUE
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who	Nullable

		updated the record	

```sql
CREATE TABLE IF NOT EXISTS supply_planning.min_max_doh (
    min_max_doh_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    group_id UUID NOT NULL,
    scenario_id UUID NOT NULL,
    month INTEGER NOT NULL,
    year INTEGER NOT NULL,
    min_doh INTEGER NOT NULL,
    max_doh INTEGER NOT NULL,
    apply_to_all_months BOOLEAN NOT NULL DEFAULT TRUE,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_minmax_group FOREIGN KEY (group_id) REFERENCES supply_planning.grouping(group_id),
    CONSTRAINT fk_minmax_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id)
);
```

17. Fluctuation allowance

Fluctuation Allowance is a configurable soft limit applied to vanning volume, ensuring that the volume does not increase by more than a specified percentage compared to the previous period.

17.1 `supply_planning.monthly_fa`

Column name	Data Type	Description	Constraints
monthly_fa_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated group	Foreign Key (`scenarios.scenario_id`), Not Null
group_id	UUID	Reference to the associated planning scenario	Foreign Key (`grouping.group_id`), Not Null
month	INTEGER	Year and month for which the FA limits apply	Not Null

year	INTEGER	Year for which monthly FA limits apply	Not Null
monthly_fa_percent	INTEGER	Allowed fluctuation percentage for the month	Not Null, Check (0 <= value <= 100)
apply_to_all_months	BOOLEAN	Shows status of toggle button on UI	Not Null, default TRUE
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Nullable
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE IF NOT EXISTS supply_planning.monthly_fa (
    monthly_fa_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    group_id UUID NOT NULL,
    month INTEGER,
    year INTEGER,
    monthly_fa_percent INTEGER NOT NULL CHECK (monthly_fa_percent >= 0 AND monthly_fa_percent <= 100),
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    apply_to_all_months BOOLEAN NOT NULL DEFAULT TRUE,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
```

```
    CONSTRAINT fk_monthly_fa_group FOREIGN KEY (group_id) REFERENCES supply_planning.grouping(group_id),
    CONSTRAINT fk_monthly_fa_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id) ON DELETE CASCADE
);
```

17.2 `supply_planning.daily_fa`

Table Name	Purpose/Role
daily_fa_mapper	Central mapping table linking scenario_id, group_id, and daily_fa_type to identify which detailed table to use.
daily_fa_weekly	Stores weekly data with fields like week_start_date, week_end_date, and week_fa.
daily_fa_monthly	Stores monthly data with fields like month, year, and month_fa.
daily_fa_custome_range	Stores data for custom date ranges with start and end dates and corresponding FA values.
daily_fa_mapper (variant)	Possibly a duplicate or related mapper table with slightly different fields (daily_fa_type instead of daily_fa).

17.2.1 `supply_planning.daily_fa_mapper`

```
CREATE TABLE supply_planning.daily_fa_mapper (
    daily_fa_mapper_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    group_id UUID NOT NULL,
    daily_fa_type TEXT NOT NULL,
    apply_to_all_days BOOLEAN NOT NULL DEFAULT TRUE,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE NOT NULL,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_daily_fa_mapper_group FOREIGN KEY (group_id) REFERENCES supply_planning.grouping(group_id),
    CONSTRAINT fk_daily_fa_mapper_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id) ON DELETE CASCADE
```

```
);
```

17.2.2 supply_planning.daily_fa

```
CREATE TABLE supply_planning.daily_fa (
    daily_fa_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    daily_fa_mapper_id UUID NOT NULL,
    daily_date UUID NOT NULL,
    daily_fa INTEGER NOT NULL,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL
DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE NOT NULL,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_daily_fa_daily_fa_mapper FOREIGN KEY
(daily_fa_mapper_id) REFERENCES
supply_planning.daily_fa_mapper(daily_fa_mapper_id)
);
```

17.2.3 supply_planning.daily_fa_weekly

```
CREATE TABLE supply_planning.daily_fa_weekly (
    daily_fa_weekly_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    daily_fa_mapper_id UUID NOT NULL,
    week_start_date DATE NOT NULL,
    week_end_date DATE NOT NULL,
    week_fa INTEGER NOT NULL,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL
DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE NOT NULL,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_daily_fa_weekly_daily_fa_mapper FOREIGN KEY
(daily_fa_mapper_id) REFERENCES
supply_planning.daily_fa_mapper(daily_fa_mapper_id)
);
```

17.2.4 supply_planning.daily_fa_custome_range

```
CREATE TABLE supply_planning.daily_fa_custome_range (
    daily_fa_custom_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    daily_fa_mapper_id UUID NOT NULL,
    custome_range_start_date DATE NOT NULL,
    custome_range_end_date DATE NOT NULL,
    custome_range_fa INTEGER NOT NULL,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL
DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE NOT NULL,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_daily_fa_custome_range_daily_fa_mapper FOREIGN KEY
(daily_fa_mapper_id) REFERENCES
supply_planning.daily_fa_mapper(daily_fa_mapper_id)
);
```

17.2.5 supply_planning.daily_fa_monthly

```
CREATE TABLE supply_planning.daily_fa_custome_range (
    daily_fa_custom_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    daily_fa_mapper_id UUID NOT NULL,
    month INTEGER NOT NULL,
    year INTEGER NOT NULL,
    month_fa INTEGER NOT NULL,
```

```sql
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE NOT NULL,
    CONSTRAINT fk_daily_fa_monthly_daily_fa_mapper FOREIGN KEY (daily_fa_mapper_id) REFERENCES supply_planning.daily_fa_mapper(daily_fa_mapper_id)
);
```

18. `supply_planning.simulation`

This table stores each simulation created under a specific scenario, mapped to its corresponding group ID.

Column name	Data Type	Description	Constraints
simulation_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated planning scenario	Foreign Key (`scenarios.scenario_id`), Not Null
group_id	UUID	Reference to the associated group	Foreign Key (`grouping.group_id`), Not Null
simulation_scenario_name	TEXT	Name of simulation scenario	Nullable
rerun_count	INTEGER	Shows the number of times the simulation has been run.	Not Null, Default 0
simulation_status	TEXT	Store status like Rejected, Approved	Not Null, Default **(TBD)**
notes	TEXT	optional note	Nullable

comments	JSONB	stores comment thread along with attachments detail	Nullable
key_input_parameters	JSONB	Contain payload that will be send to DS Lambda	Nullable
activity_log			
is_active	BOOLEAN	Shows simulation is active or not	Not Null, Default True
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Not Null
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE supply_planning.simulation (
    simulation_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    group_id UUID NOT NULL,
    simulation_scenario_name TEXT,
    rerun_count INTEGER NOT NULL DEFAULT 0,
    simulation_status TEXT NOT NULL,
    note TEXT,
    comments JSONB,
    key_input_parameters JSONB,
    activity_log JSONB,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
```

```sql
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP,
    last_updated TIMESTAMP WITHOUT TIME ZONE NOT NULL,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_simulation_group FOREIGN KEY (group_id) REFERENCES supply_planning.grouping(group_id),
    CONSTRAINT fk_simulation_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id) ON DELETE CASCADE
);
```

19. `supply_planning.scenario_step_status`

This table stores the status of each input and is used to support the UI sidebar.

Column name	Data Type	Description	Constraints
scenario_step_status_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated simulation	Foreign Key (`scenarios.scenario_id`), Not Null
input_step_type	TEXT	e.g., Line level or vanning canter inputs	Not Null
input_step_name	TEXT	Model change date, Namc allocation plan	Not Null
status	TEXT	input status	Nullable
updated_by	TEXT	Username who updated inputs	Nullable
last_updated	TIMESTAMP	Timestamp when the record was last updated	Not Null
created_by	TEXT	Email ID of the user who created the record	Not Null

updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE supply_planning.scenario_step_status (
    scenario_step_status_id  UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    input_step_type TEXT,
    input_step_name TEXT,
    status TEXT,
    updated_by TEXT,
    last_updated TIMESTAMP WITHOUT TIME ZONE NOT NULL,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_scenario_step_status_scenario FOREIGN KEY (scenario_id) REFERENCES supply_planning.scenarios(scenario_id),
    CONSTRAINT scenario_step_status_scenario_id_input_step_name_key
        UNIQUE (scenario_id, input_step_name)
);
```

19. `supply_planning.file_upload_log`

This table will be used by APIs as well as DE team to log processing status for file upload.

Column name	Data Type	Description	Constraints
job_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated scenario	Foreign Key (`scenarios.scenario_id`), Not Null
step_name	TEXT	Name of the processing step (e.g., namc production calendar, vanninf lead time)	Not Null
execution_status	TEXT	Current execution status	Nullable

		of the job (e.g., Success, Failed)	
error_message	TEXT	Detailed error message captured if the job execution fails	Nullable
file_path	TEXT	S3 file path where file is stored	Not Null
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Not Null
created_by	TEXT	Email ID of the user who created the record	Not Null
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE supply_planning.file_upload_log (
    job_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    step_name TEXT NOT NULL,
    execution_status TEXT NOT NULL,
    error_message TEXT NOT NULL,
    file_path TEXT NOT NULL,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_file_upload_log_scenario FOREIGN KEY (scenario_id)
REFERENCES supply_planning.scenarios(scenario_id)
);
```

20. `supply_planning.pre_population_log`

This table will be used by APIs as well as DE team to log pre-population processing step.

Column name	Data Type	Description	Constraints
pre_populate_id	UUID	Unique identifier for each record	Primary Key
scenario_id	UUID	Reference to the associated scenario	Foreign Key (`scenarios.scenario_id`), Not Null
input_type	TEXT	Name of the input (e.g., shipping pattern, model change date)	Not Null
execution_status	TEXT	Current execution status of the job (e.g., Success, Failed)	Not Null
error_message	TEXT	Detailed error message captured if the job execution fails	Nullable
created_date_timestamp	TIMESTAMP	Timestamp when the record was created	Not Null, Default current timestamp
last_updated	TIMESTAMP	Timestamp when the record was last updated	Not Null
created_by	TEXT	Email ID of the user who	Not Null

		created the record	
updated_by	TEXT	Email ID of the user who updated the record	Nullable

```sql
CREATE TABLE supply_planning.pre_population_log (
    pre_populate_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario_id UUID NOT NULL,
    input_type TEXT NOT NULL,
    execution_status TEXT NOT NULL,
    error_message TEXT NOT NULL,
    created_date_timestamp TIMESTAMP WITHOUT TIME ZONE NOT NULL,
    last_updated TIMESTAMP WITHOUT TIME ZONE,
    created_by TEXT NOT NULL,
    updated_by TEXT,
    CONSTRAINT fk_file_upload_log_scenario FOREIGN KEY (scenario_id)
REFERENCES supply_planning.scenarios(scenario_id)
);
```

21. `supply_planning.daily_rundown`

This represents the estimated output structure based on the previous system. The current model output is under review, and the structure will be updated once it is finalized.
This is added just for reference.

Column name	Data Type	Constraints
id	UUID	Primary Key
simulation_id	UUID	Foreign Key (`simulation.simulation_id`), Not Null
scenario_id	UUID	Foreign Key (`scenarios.scenario_id`), Not Null
group_id	UUID	Foreign Key (`grouping.group_id`), Not Null
importer	TEXT	Not Null

rundown_key	TEXT	Not Null
variation_version	TEXT	Nullable
ssn_no	TEXT	Nullable
id_line	TEXT	Nullable
exporter	TEXT	Not Null
order_date	DATE	Nullable
date	DATE	Not Null
total	INTEGER	Not Null, Default 0
prod	INTEGER	Not Null, Default 0
ttl_ba	TEXT	Nullable
occur_ba	TEXT	Nullable
reflect_ba	TEXT	Nullable
local_stock	INTEGER	Not Null, default 0
stock_days	DECIMAL(5,2)	Not Null, default 0
unload	INTEGER	Not Null, default 0
transit	INTEGER	Not Null, default 0
loading	INTEGER	Not Null, default 0
port_stock	INTEGER	Not Null, default 0
van_pack	INTEGER	Not Null, default 0
lot_size	INTEGER	Not Null, default 0
numberof_lot	INTEGER	Not Null, default 0
total_stock	INTEGER	Not Null, default 0